

Samhita Pal

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Education	Ph.D. in Statistics	Jan 2021 - May 2025
	North Carolina State University (NCSU), USA Thesis: Bayesian Inference in High-dimensional Regression Models using Sparse Projection Posterior. Advisor: Subhashis Ghoshal GPA: 4.00	
	M.Sc. Statistics	August 2018 - October 2020
	Department of Statistics, University of Calcutta Course rank: 2	
	B.Sc. Statistics (Hons.)	July 2015 - May 2018
	St. Xavier's College (Autonomous), Kolkata, India Course rank: 1	
Experience	Postdoctoral Associate in Biostatistics	Jun 2025 - present
	Vanderbilt University Medical Center (VUMC), USA Advisors: Amir Asiaee (VUMC) and Jared Huling (University of Minnesota)	
	North Carolina State University, USA	August 2024 - May 2025
	Graduate Teaching Assistant Department of Statistics	
	AstraZeneca Pharmaceuticals, USA	June 2024 - August 2024
	Summer Biometric Intern Oncology Department	
	North Carolina State University, USA	January 2022 - May 2024
	Graduate Research Assistant Department of Statistics	
	North Carolina State University, USA	January 2021 - December 2021
	Graduate Teaching Assistant Department of Statistics	
	National University of Singapore, Singapore	August 2019
	Graduate Program Workshop on Statistical Data Integration by Institute of Mathematical Sciences	
	Indian Institute of Science Education and Research, India	May - July 2019
	Summer Research Internship Department of Mathematics and Statistics	

Honors & Awards

- **Finalist for Savage Award** for dissertation at ISBA 2026.
- **Fall 2024 NC State Graduate Student Association Award for Travel and Conferences, Student Conference Award.**
- **IMS 2024 ICSDS Student Paper Competition Travel Award.**
- **WSDS 2023 Travel Award.**
- **National Science Foundation Travel Award** (for New Frontiers in Reliability and Risk Analysis Symposium, 2023).
- **IISA 2023 Student Paper Competition Travel Award.**
- **B.B. Bhattacharyya Graduate Fund for Excellence Award** (nominated, 2022; won, 2023).
- **Harshbarger Student Travel Award** (for SRCOS Conference, 2022).
- **Gertrude M.Cox Achievement Award for Outstanding PhD Qualifier Exam** (won, 2022).
- **Scholarship for Higher Education (SHE): Innovation in Science Pursuit for Inspired Research (INSPIRE)** (from Department of Science and Technology, Ministry of Science and Technology, Government of India, 2015 - 2020)
- **Third Prize for Poster** (National Seminar for Sample Surveys, India, 2019).
- **Second rank in M.Sc Statistics** (Department of Statistics, University of Calcutta, India, 2020)
- **Arts and Science Award for securing First rank in Bachelor of Science in Statistics** (Batch: 2015 - 2018, St. Xavier's College (Autonomous), Kolkata).
- **First Prize for top score in school.**
- **First Rank in school along with under-25 rank in state** (Higher Secondary Examination, West Bengal, India, 2015).
- **First Rank in school along with under-25 rank in state** (Secondary Examination, West Bengal, India, 2013).
- **Honorary student membership at Mu Sigma Rho.**

Published

Bayesian High-dimensional Grouped-regression using Sparse Projection-posterior

Samhita Pal¹, Subhashis Ghoshal¹

¹NCSU

Statistica Sinica, 2026.

Sharp Bounds for Treatment Effect Generalization under Outcome Distribution Shift

Amir Asiaee¹, **Samhita Pal**¹, Cole Beck¹ and Jared D. Huling²

¹VUMC, ²University of Minnesota.

Accepted at CLear 2026.

Efficacy Analysis in Clinical Trials: A Comprehensive Review of Statistical and Machine Learning Approaches

Dhrubajyoti Ghosh¹, **Samhita Pal**²

¹Duke Biostatistics, ²NCSU

Accepted at NEJSDS, 2026.

Projection-posterior for variable selection: Weak limit and coverage

Samhita Pal¹, Subhashis Ghoshal¹

¹NCSU

Electronic Journal of Statistics, 2025.

Ensemble survival analysis for preclinical cognitive decline prediction in Alzheimer's disease using longitudinal biomarkers

Dhrubajyoti Ghosh¹, **Samhita Pal**², Michael Lutz³, Sheng Luo¹

¹Duke Biostatistics, ²NCSU, ³Duke Neurology

Journal of Alzheimer's Disease, 2025.

Discovering Candidate Genes Regulated by GWAS Signals in Cis & Trans

Samhita Pal¹, Xinge Jessie Jeng¹

¹NCSU

Statistics in Biosciences, 2024.

Revised

Penalized FCI for Causal Structure Learning in a Sparse DAG for Biomarker Discovery in Parkinson's Disease

Samhita Pal¹, Dhrubajyoti Ghosh², Shu Yang¹

¹NCSU, ²Duke Biostatistics

Revision submitted to Annals of Applied Statistics.

Bayesian High-dimensional Linear Regression with Sparse Projection-posterior

Samhita Pal¹, Subhashis Ghoshal¹

¹NCSU

Revision submitted to Bayesian Analysis

Alpha-allocation based on Utility and Likelihood in Multi-arm Trials

Samhita Pal¹, Xiaoling Wu²

¹NCSU, ²AstraZeneca

Revision submitted to Biometrical Journal.

Submitted

Improving Precision of RCT-Based CATE Estimation using Data Borrowing with Double Calibration

Amir Asiaee¹, Chiara Di Gravio², Cole Beck¹, Yuting Mei¹, **Samhita Pal**¹, and Jared D. Huling³

¹VUMC, ²Imperial College London, ³University of Minnesota.

Submitted to JMLR.

Improving RCT-Based CATE Estimation Under Covariate Mismatch via Double Calibration

Samhita Pal¹, Amir Asiaee¹, Jared Huling²

¹VUMC, ²University of Minnesota

Submitted to JASA.

Improving RCT-Based Treatment Effect Estimation Under Covariate Mismatch via Calibrated Alignment

Samhita Pal¹, Amir Asiaee¹ *Submitted to UAI 2026*

Network-aware IV Regression for Causal Node Discovery and Estimation

Samhita Pal¹, Dhrubajyoti Ghosh² ¹VUMC, ²Duke Biostatistics

Submitted to JASA.

DAG DECORation: Continuous Optimization for Structure Learning under Hidden Confounding

Samhita Pal¹, James O’quinn², Kaveh³ Aryan, Heather Pua¹, James P. Long⁴, Amir Asiaee¹

¹VUMC, ²VU, ³King’s College, London, ⁴MDACC

Submitted to ICML 2026.

Causal Discovery with Mixed Latent Confounding via Precision Decomposition

Amir Asiaee¹, **Samhita Pal**¹, James O’quinn², James P. Long³

¹VUMC, ²JHU, ³MDACC

Submitted to UAI 2026.

Representation Learning for Causal Inference in Multimodal Repeated Cross-Sectional Surveys with Block-missingness

Dhrubajyoti Ghosh¹, **Samhita Pal**²

¹Kennesaw State University, ²VUMC

Submitted to DATA 2026.

A Bayesian Model for Peer Assessors’ Reputation System

Parvez Rashid, **Samhita Pal**, Vinay Kumar Samudrala, Rahul Yedida and Edward Gehringer

Submitted to EDM 2026.

Air Pollution and Public Health: A Lagged Spatio-Temporal Analysis of COPD and Stroke across U.S. Cities

Samhita Pal¹, Dilshad Imon², Dhrubajyoti Ghosh³

¹NCSU, ²UNC, ³Duke

Submitted to Journal of Data Science

Working

Hierarchical Projection for Adaptive Knowledge Transfer

Samhita Pal¹, Tian Gu²

¹VUMC, ²Columbia University

Heterogeneous Effects of Continuous Treatments via Conditional Incremental Effects

Samhita Pal¹, Jared Huling²

¹VUMC, ²University of Minnesota

Social Sensors in Epidemiological Networks via Graph Eigenvectors

Shubhajit Sen¹, **Samhita Pal**¹ and Srijan Sengupta¹

¹NCSU

Conference & Presentations

Finalist for the ISBA Savage Award (2026), Best Dissertation in the Theory and Methods category.

ICSIDS: International Conference on Statistics and Data Science (2024), Student Paper Award Session.

CFE-CMStatistics: 18th International Joint Conference on Computational and Financial Econometrics and Computational and Methodological Statistics (2024), Invited Talk.

Calcutta Statistical Association Triennial Symposium (2024), **Invited Talk.**

Graduate Student Research Symposium, NCSU (2024), **Poster Presentation.**

WSDS: Women in Statistics and Data Science (2023), **Speed Talk and Hybrid Poster Session.**

New Frontiers in Reliability and Risk Analysis Symposium: A Tribute to Nozer D. Singpurwalla (2023), **Invited Poster Session.**

JSM: Joint Statistical Meetings (2023), **Poster Presentation.**

IISA: International Indian Statistical Association (2023), **Student Paper Competition** (Theory and Probability division).

SRCOS: Southern Regional Council On Statistics (2022), **Poster Presentation.**

National Seminar for Sample Survey, University of Calcutta (2019), **Poster Presentation.**

Teaching Experiences

ST 558 — Data Science for Statisticians (Fall 2024) *Grader*
ST 563 — Introduction to Statistical Learning (Fall 2024) *Grader*
ST 503 — Fundamentals of Linear Models and Regression (Summer 2023) *Grader*
ST 312 — Introduction to Statistics II (Fall 2021): *Grader*
ST 311 — Introduction to Statistics (Summer 2021) *Grader*
ST 312 — Introduction to Statistics II (Spring 2021) *Lab Instructor*

Research Interests

High-dimensional Models
Bayesian Modeling and Inference
Causal Inference
Instrumental Variable Regression
Transfer Learning

Computing Skills

R (Proficient)
SAS (Intermediate)
Python (Elementary)

R packages

sparseProj <https://github.com/SamhitaPal3/sparseProj>
credInt <https://github.com/SamhitaPal3/credInt>

Course Projects

“A Simulation Based Study of Covariate-assisted Network Clustering”
PhD course project, Spring 2022

“Modeling Interaction Effects in eQTL Mapping”
PhD course project, Fall 2021

“On Bivariate Penalized Smoothing for Irregularly Sampled Data”
PhD Course Project, Spring 2021

“Predicting Student Success using Multivariate Response Models”

PhD Course Project, Spring 2021

“A Study of Count Regression in Presence of Outliers”
University of Calcutta, India, 2020

Consulting

“Land-based Weather Analysis: A Spatio-Temporal Model”; collaborated with Shana McDowell, PhD student, Dr. Daniela Jones, Professor; Department of Biological and Agricultural Engineering, NC State University, USA.

“Head Injury in Children: A Two Years Study”; collaborated with Dr. Sandip Pal, Associate Professor; Department of Neurosurgery, Calcutta Medical College, India.

Paper Reviews

- Calcutta Statistical Association Bulletin
- Annals of Statistics
- Asian Research Journal of Mathematics
- CLear Conference 2026
- Bayesian Analysis
- AISTATS CauScale 2026
- BMC Medical Informatics and Decision Making
- Frontiers in Applied Mathematics and Statistics
- Asian Journal of Probability and Statistics
- NeurIPS 2026

Languages known

English, Bengali, Hindi, Spanish (elementary)

Positions of Responsibility

- Organized Invited Session titled “**Advances in statistical modeling and machine learning for complex data**” at the 2025 CFE-CMStatistics.
- Founder President of Triangle Bengali Association, an NC State University organization for the Bengali-speaking community. (2022 - 2024).
- International Students Representation of Statistical Graduate Students Association, NC State University (2022 - 2024) *Organized 4 events.*
- Volunteer at Office of International Services, NC State University (2022 - 2024).

Other Activities

- Completed upto 4th year in Indian classical dance forms- Bharatnatyam and Manipuri from Pracheen Kala Kendra, Chandigarh, India.
- Completed upto 4th year in Fine Arts from Sarbabharatiya Sangeet-O-Sanskriti Parishad, recognised by West Bengal State Academy of Dance, Drama, Music Fine Arts and Raja Mansingh Tomar Music Arts University, Gwalior, MP.
- Ukulele, Classical Music, Badminton